Facial Pain Classification Using Deep Neural Networks for Clinical Decision Support

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Abstract

Accurate pain assessment is a critical challenge in healthcare, particularly for patients who are unable to communicate effectively, such as infants, elderly individuals, and patients with neurological impairments. Traditional pain assessment methods rely heavily on subjective observation, which may lead to inconsistent and unreliable evaluations.

In this project, we develop a deep learning-based system for automatic facial pain intensity classification using image data. The task is formulated as a multi-class classification problem with four categories: No Pain, Mild Pain, Moderate Pain, and Severe Pain. To address this problem, multiple deep learning models are implemented and compared, including a baseline Convolutional Neural Network (CNN) and several transfer learning architectures such as VGG16, ResNet50, and EfficientNet-B3. In addition, a dual-input architecture that combines full-face and mouth-region features is designed to capture both global and local facial cues.

Experimental results demonstrate that transfer learning significantly improves classification performance compared to the baseline model. These findings highlight the importance of pretrained feature extraction in complex visual classification tasks. Overall, the proposed system shows strong potential for assisting clinical decision-making in pain assessment scenarios. The proposed dual-input architecture further demonstrates the effectiveness of combining global and localized facial features for fine-grained pain classification.

1. Introduction

Pain assessment plays a fundamental role in clinical care and treatment planning. However, in many real-world situations, patients are unable to clearly express their pain levels, which introduces uncertainty and variability in diagnosis. This limitation motivates the development of automated systems capable of objectively estimating pain intensity using observable signals such as facial expressions.

Facial expressions are known to reflect emotional and physical states, including pain. Facial analysis is a promising direction because pain often affects muscles around the eyes, eyebrows, cheeks, and mouth. Lighting conditions, head pose, image quality, age differences, and subject-specific facial structure can make the task challenging. Nevertheless, distinguishing between different levels of pain based on facial cues is a challenging problem. The differences between categories such as mild and moderate pain are often subtle and difficult to capture, even for

human observers. Therefore, this task requires models capable of extracting fine-grained visual features.

Recent advances in deep learning have demonstrated strong performance in image classification tasks. In this project, we leveraged these advances to build a facial pain classification system. The objective is not only to develop a working model, but also to systematically compare different architectures and understand their strengths and limitations. The proposed dual-input architecture further demonstrates the benefit of combining global and localized facial features for fine-grained classification tasks.

2. Problem Statement

The objective of this project is to design a multi-class classification system that predicts pain intensity levels from facial images. Each image is categorized into one of four classes: No Pain, Mild Pain, Moderate Pain, and Severe Pain. The model is expected to achieve strong predictive performance, generalize well to unseen data, and maintain balanced performance across all classes.

From a machine learning perspective, this is a four-class supervised classification problem. Given an RGB facial image, the system must predict the corresponding pain-intensity label. The main technical goals of the project are:

- Build a complete end-to-end deep learning pipeline for pain classification.
- Establish a baseline using a custom CNN.
- Improve performance using stronger architecture and training strategies.
- Evaluate the system using balanced metrics appropriate for multi-class classification.
- Prepare the model for simple web deployment and optional visual interpretation.

3. Related Work

Deep learning has become the dominant approach for image classification due to the ability of convolutional neural networks (CNNs) to learn hierarchical visual features directly from raw data. In healthcare applications, CNN-based models have been widely used for tasks such as diagnosis support, facial analysis, and expression recognition. In this project, we follow a similar approach by first implementing a baseline CNN and then applying transfer learning using pretrained architectures such as ResNet50, VGG16, and EfficientNet. These models leverage knowledge learned from large-scale datasets and typically achieve improved performance when task-specific data is limited.

Prior work in both academic research and student projects demonstrates a consistent trend: while baseline models provide a useful reference point, performance improves significantly through transfer learning, data augmentation, and stronger regularization techniques. For

example, in a Human Action Recognition study, multiple architectures were compared, and pretrained models consistently outperformed a custom CNN baseline on a complex visual classification task. These findings highlight the importance of deep pretrained backbones for extracting meaningful and discriminative features. Additionally, prior project reports emphasize that careful preprocessing, balanced evaluation, and systematic comparison of models are essential components of a successful deep learning pipeline. Our work follows this same iterative strategy, starting from a baseline and progressively improving model performance.

Beyond general image classification, facial pain detection has evolved significantly over time. Early work by Cohn et al. (2009) demonstrated that pain can be identified through specific facial action units (AUs), particularly in regions such as the brow, eyes, and nasolabial area. Their study established that pain expressions are localized rather than uniformly distributed across the face. However, they also highlighted a key limitation: pain-related expressions often overlap with other emotions such as smiling, making accurate classification difficult. This challenge remains relevant in modern deep learning systems, where subtle and ambiguous expressions can still lead to misclassification.

More recent research has shifted toward deep learning approaches that incorporate both spatial and temporal information. For example, Tan et al. (2025) proposed a spatial-temporal attention long short-term memory (STA-LSTM) model that analyzes facial landmark sequences extracted from clinical video data. By combining spatial attention, which focuses on important facial regions, with temporal attention, which emphasizes informative frames, their approach captures the dynamic nature of pain expressions. This represents a significant improvement over static image-based methods and achieves strong performance in both binary and multi-level pain classification tasks.

In addition to methodological advancements, recent studies have increasingly focused on real-world deployment of pain detection systems. For instance, Almarri et al. (2023) proposed a deep learning–based framework for pain-level detection designed for patient-centric healthcare applications in the UAE. Their work evaluates multiple deep learning architectures within a complete system pipeline, including preprocessing, model training, and deployment considerations. The study highlights the importance of scalability, usability, and real-time inference, emphasizing that successful systems must go beyond model accuracy to meet practical clinical requirements. These findings align with our approach, where multiple architectures are compared and the final system is designed with deployment in mind through a user-friendly interface.

Similarly, systems such as PainChek demonstrate how automated facial analysis can be integrated into clinical workflows by combining facial recognition with additional observational inputs, such as voice, behavior, and physiological indicators. While such systems show promising

results, they also reveal ongoing challenges, including limited validation datasets, potential demographic bias, and concerns related to privacy and data security. These challenges emphasize the importance of developing models that are not only accurate but also fair, interpretable, and reliable in real-world settings.

Overall, the literature shows a clear progression in facial pain detection research, moving from handcrafted feature-based methods to deep learning models with attention mechanisms, and more recently to multimodal and deployment-oriented systems. Despite these advancements, significant challenges remain, particularly in distinguishing between visually similar expressions and achieving robust performance across diverse conditions. In contrast to multimodal and temporally-aware approaches, our project focuses on CNN-based classification using static facial images. While this enables effective model development and comparison, it may limit the ability to capture temporal dynamics. These findings suggest that future work could benefit from incorporating sequential information, multimodal inputs, and more clinically accurate datasets to further improve performance and real-world applicability.

4. Dataset Description

The dataset used in this project is the Balanced RGB Facial Expression dataset, which is based on AffectNet and consists of real-world facial images collected under diverse conditions. Unlike traditional laboratory datasets that often use grayscale images, this dataset contains genuine RGB images, where each color channel provides distinct visual information. This characteristic allows deep learning models to capture richer and more realistic facial features. Additionally, the dataset is pre-balanced across eight emotion categories and is available in multiple scales (1×, 2×, and 3×), making it suitable for robust model training.

The original dataset contains emotion-based labels, including *neutral, happy, sad, fear, anger, disgust, surprise,* and *contempt*. To adapt the dataset for the pain classification task, an automatic relabeling strategy was applied to map these emotion categories into four pain intensity levels. Specifically, *neutral* and *happy* expressions were mapped to *no pain*, while *sad* and *contempt* were mapped to *mild pain*. Expressions associated with visible tension, such as *fear*, were mapped to *moderate pain*, and more intense expressions, including *anger, disgust,* and *surprise*, were mapped to *severe pain*.

The inclusion of the *contempt* class is particularly important, as it represents subtle unilateral facial tension and was therefore assigned to the *mild pain* category. This mapping process was implemented automatically within the preprocessing pipeline, ensuring consistency across all data splits.

This approach enables the dataset to approximate pain-related facial patterns while maintaining a balanced class distribution. However, since the labels are derived from emotional expressions

rather than clinical pain annotations, the resulting model may capture general facial intensity rather than true pain-specific features.

While the balanced nature of the dataset helps reduce bias, the visual differences between some classes remain subtle, making the classification task inherently challenging.

The properties of this dataset also support the design of the dual-input architecture. Since the images are high-quality RGB photographs, they preserve detailed facial features, particularly in the mouth region, which plays a key role in expressing pain. Additionally, the relabeling strategy maps several subtle expressions, such as sadness and contempt, to mild pain, where facial changes are often localized rather than global.

Motivated by this, the dual-input model combines full-face images with cropped mouth-region inputs to capture both global context and localized features. This design aligns the dataset characteristics with the model architecture, improving the ability to detect subtle differences between pain levels.

5. Data Preprocessing

To ensure consistency and improve model performance, a series of preprocessing steps were applied. All images were resized to a fixed resolution to match the input requirements of the neural networks. Pixel values were normalized to stabilize training and improve convergence.

In addition, face detection and cropping were applied to focus the model on the most relevant regions of the image. This step reduces background noise and ensures that the input primarily contains facial features associated with pain expression.

Data augmentation techniques were applied to the training set to increase diversity and reduce overfitting. These techniques include horizontal flipping, small rotations, and brightness or contrast adjustments. Augmentation improves the model's ability to generalize by exposing it to variations in pose, lighting, and facial appearance.

Finally, the dataset was divided into training, validation, and test sets. The training set was used for learning model parameters, the validation set for model selection and hyperparameter tuning, and the test set for final performance evaluation.

6. Methodology

To address the problem, a multi-model approach is adopted. Rather than relying on a single architecture, several models are implemented and compared to identify the most effective solution.

A baseline CNN is first developed to establish a reference performance. This model is trained from scratch and provides insight into how well a simple architecture can capture facial features related to pain.

Subsequently, transfer learning models are explored. Pretrained architectures such as VGG16, ResNet50, and EfficientNet-B3 are used, leveraging features learned from large-scale datasets. These models are fine-tuned for the specific task of pain classification.

In addition to single-input models, a dual-input architecture is designed to improve performance. This model processes both the full facial image and a cropped mouth region, combining global and local features for classification. Since the mouth region plays a significant role in expressing pain, incorporating this information explicitly is expected to enhance model performance.

7. Model Architectures

The baseline model is a custom CNN consisting of convolutional layers, activation functions, pooling layers, and a fully connected classifier. While simple, it provides an important benchmark.

VGG16 is a deep convolutional network known for its strong feature extraction capabilities. ResNet50 introduces residual connections, enabling deeper and more stable training. EfficientNet-B3 improves performance through efficient scaling of network dimensions.

The dual-input model represents the most advanced architecture in this project. It uses two parallel branches to process the face and mouth images separately, followed by feature fusion and classification. This design allows the model to focus on both overall facial structure and localized expressions of pain.

8. Training Strategy

All models were trained using cross-entropy loss and the Adam optimizer. Training was conducted with a batch size of approximately 32 over 50 epochs. Early stopping is used to prevent overfitting, and model checkpointing ensures that the best-performing model is retained. Regularization techniques such as dropout are applied to improve generalization.

9. Evaluation Metrics

Model performance is evaluated using accuracy, precision, recall, and F1-score (macro). The macro-averaged F1-score is particularly important because it treats all classes equally, providing a balanced evaluation.

A confusion matrix is also used to analyze classification errors and identify patterns of misclassification.

10. Experimental Results and Visual Analysis

This section presents a comprehensive evaluation of the implemented models using both quantitative metrics and visual analysis. In addition to reporting accuracy and F1-score, we include confusion matrices, training curves, and Grad-CAM visualizations to better understand model behavior and performance.

10.1 Model Performance Comparison

As shown in Figure 1, the performance of all models is compared in terms of accuracy, F1-score, precision, and recall. The results clearly show that transfer learning models significantly outperform the baseline custom CNN. The custom CNN achieves relatively low performance, confirming that training from scratch is insufficient for capturing complex facial features.

Among the evaluated models, VGG16 achieves the best overall performance, with the highest F1-score and accuracy. EfficientNet-B3 and ResNet50 also demonstrate strong and consistent performance, indicating that pretrained deep architectures are highly effective for this task.

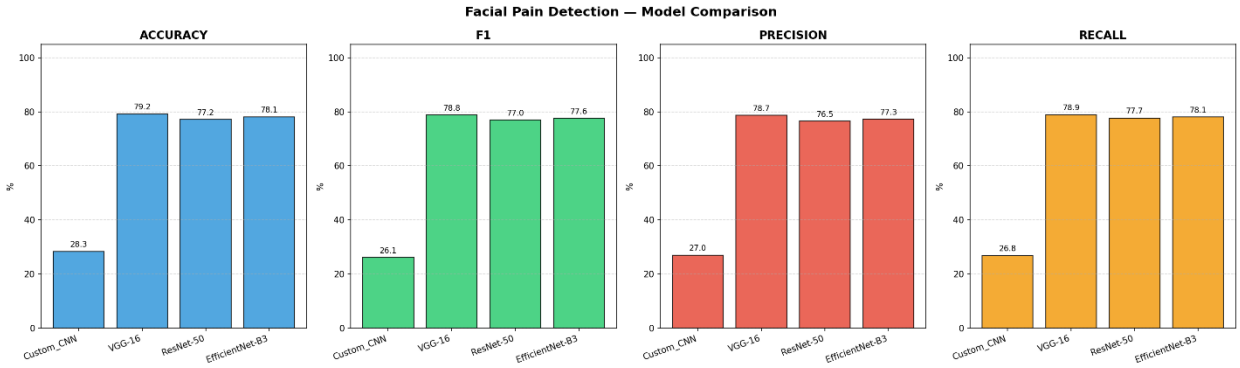


Figure 1: Overall performance comparison of all models

10.2 Confusion Matrix Analysis

The confusion matrix for the custom CNN reveals poor classification performance across all classes. The model frequently misclassifies samples, particularly confusing mild, moderate, and no pain categories. This indicates that the baseline model is unable to capture subtle facial differences required for accurate pain classification.

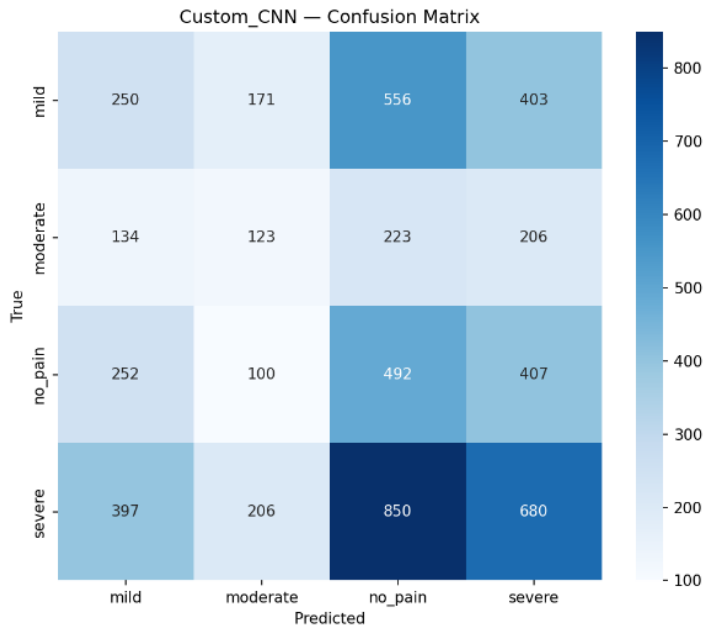


Figure 2: Confusion matrix for Custom CNN

The VGG16 model shows strong classification performance, with most predictions concentrated along the diagonal. This indicates that the model correctly classifies the majority of samples. The model performs particularly well on the *severe* and *no pain* classes, while minor confusion remains between *mild* and *moderate* categories.

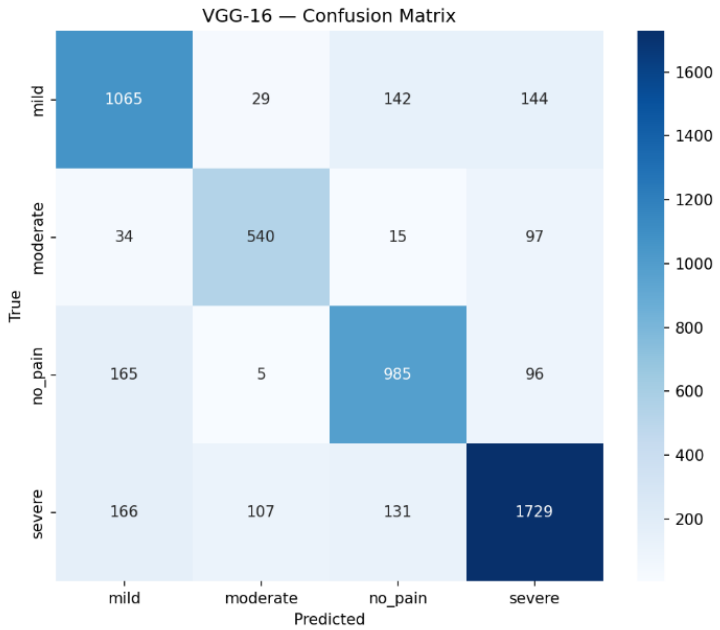


Figure 3: Confusion matrix for VGG16

ResNet50 demonstrates significant improvement over the baseline model, with strong diagonal dominance in the confusion matrix. However, similar to VGG16, some misclassification persists between adjacent classes such as *mild* and *moderate*, reflecting the inherent difficulty of distinguishing subtle expressions.

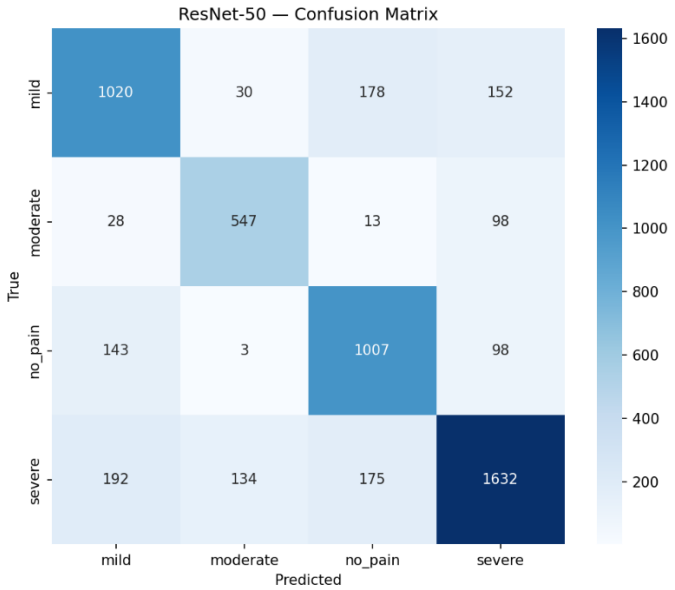


Figure 4: Confusion matrix for ResNet50

EfficientNet-B3 achieves performance comparable to VGG16, with high accuracy across all classes. The confusion matrix shows balanced performance and fewer misclassifications overall, indicating that the model generalizes well to unseen data.

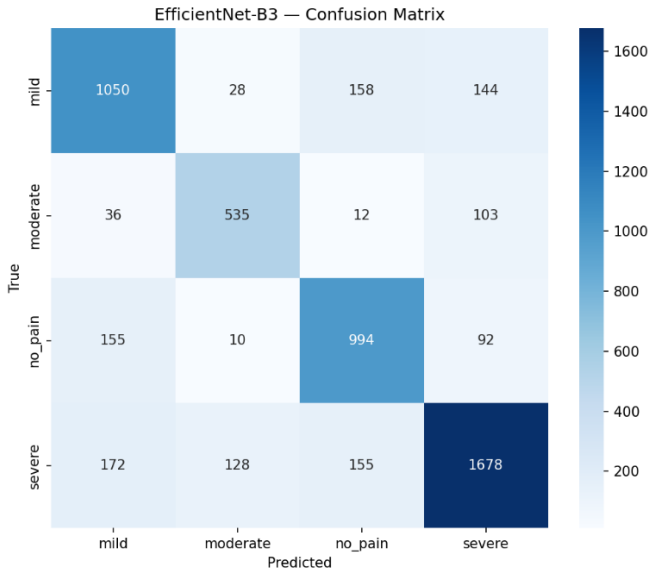


Figure 5: Confusion matrix for EfficientNet-B3

10.3 Training Behavior Analysis

Custom CNN shows weak training behavior compared to the transfer learning models. As shown in Figure 6, both training and validation accuracy remain low throughout the training process. The model fails to capture meaningful patterns from the data, resulting in poor performance.

Additionally, the training and validation curves remain close to each other, indicating that the model is not overfitting but rather underfitting. This suggests that the model lacks sufficient capacity and would benefit significantly from more advanced architecture.

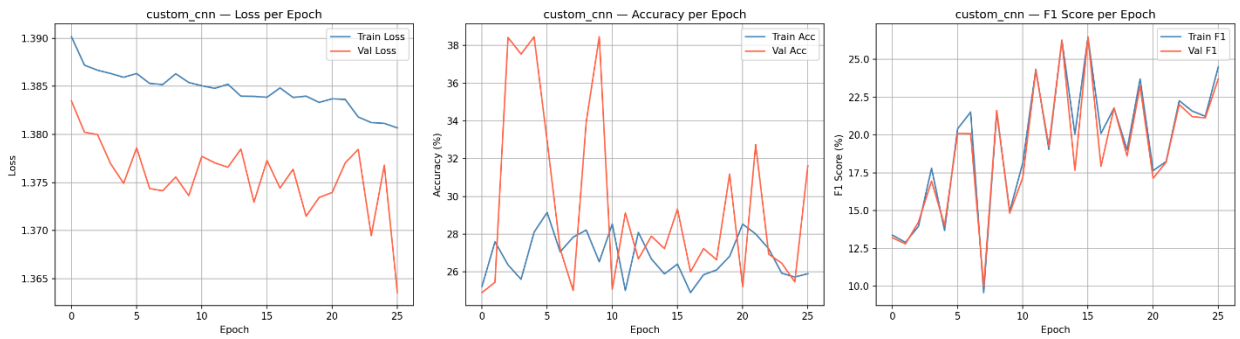


Figure 6: Training curves for Custom CNN model

VGG16 shows steady improvement in both training and validation performance. As shown in Figure 7, accuracy and F1-score increase consistently over epochs, while validation performance stabilizes after several epochs, indicating good convergence. A small gap between training and validation curves suggests mild overfitting, which is expected in deep learning models.

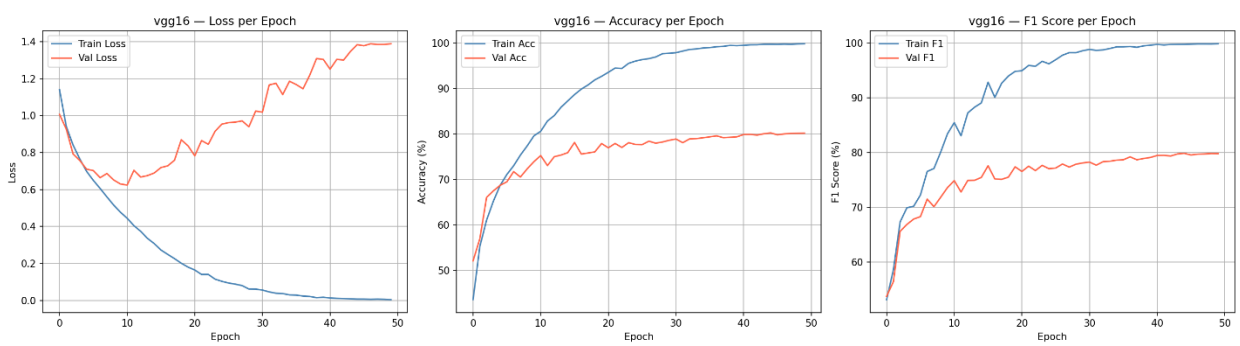


Figure 7: Training curves for VGG16

ResNet50 also demonstrates stable learning behavior during training. As shown in Figure 8, training performance improves consistently, while validation performance increases and then stabilizes. Although ResNet50 achieves strong results, the gap between training and validation curves indicates mild overfitting. This suggests that while residual connections enable deeper feature learning, additional regularization could further improve generalization.

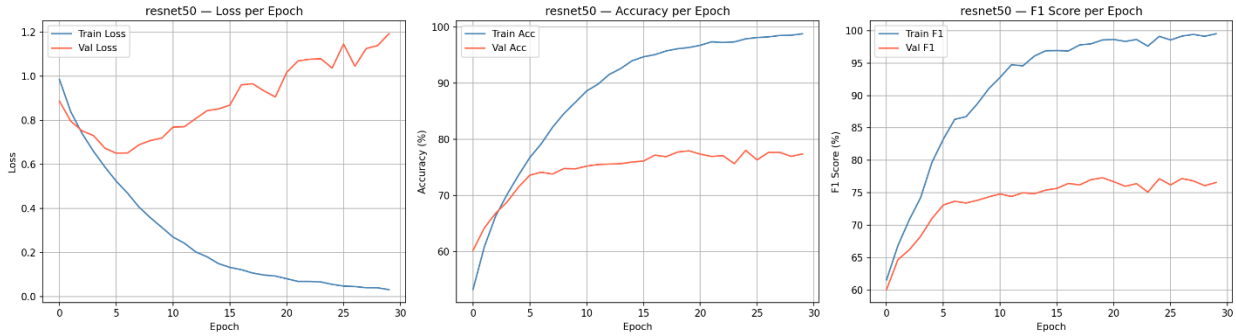


Figure 8: Training curves for ResNet50

EfficientNet-B3 demonstrates rapid convergence and strong learning capability. As shown in Figure 9, training accuracy increases significantly, while validation performance stabilizes at a high level. However, a noticeable gap between training and validation curves suggests some degree of overfitting. This indicates that additional regularization techniques could further improve performance.

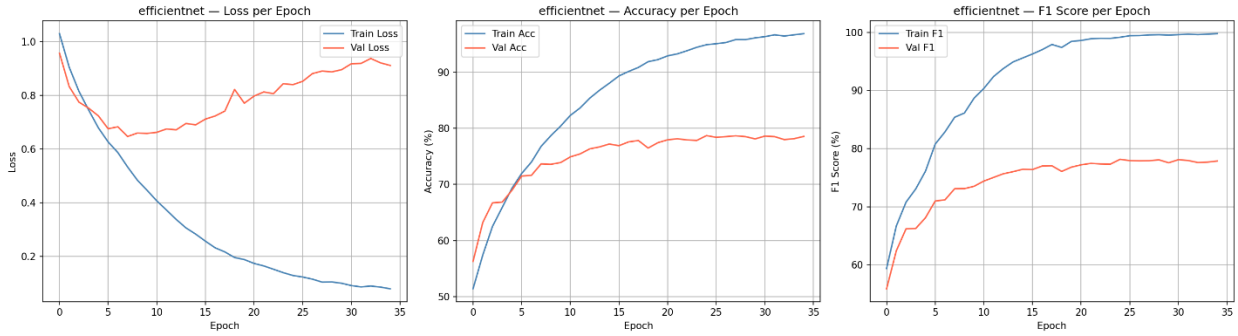


Figure 9: Training curves for EfficientNet-B3

The dual-input model shows consistent improvement in both training and validation performance. As shown in Figure 10, the validation curves are more stable compared to single-input models, suggesting improved generalization. This indicates that combining full-face and mouth-region inputs helps the model better capture both global and localized features of pain expression.

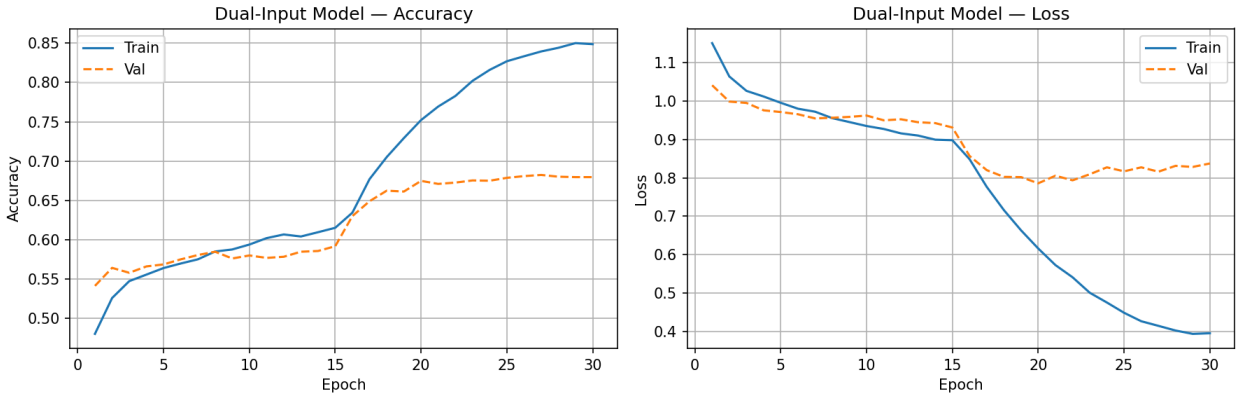


Figure 10: Training curves for Dual-Input Model

10.4 Model Interpretability using Grad-CAM

Grad-CAM visualizations provide insight into which regions of the image contribute most to the model's predictions. In the dual-input model, attention is more strongly focused on the mouth region, which is a key area for pain expression. This demonstrates that incorporating localized inputs helps the model capture meaningful facial features.

This suggests that incorporating localized facial information provides additional discriminative power, particularly for distinguishing between subtle pain levels such as mild and moderate.

Grad-CAM analysis across all models further supports the validity of the learned features. The visualizations consistently highlight regions around the eyes and mouth, which are known to be critical for expressing pain. This alignment between model attention and human understanding increases confidence in the model's predictions.

This section presents additional examples of model predictions and Grad-CAM visualizations across different architectures, including EfficientNet, ResNet50, and Dual-input. These supplementary results provide further evidence of the model's behavior and demonstrate consistent attention to key facial regions, particularly the mouth and lower facial area, which are strongly associated with pain-related expressions.

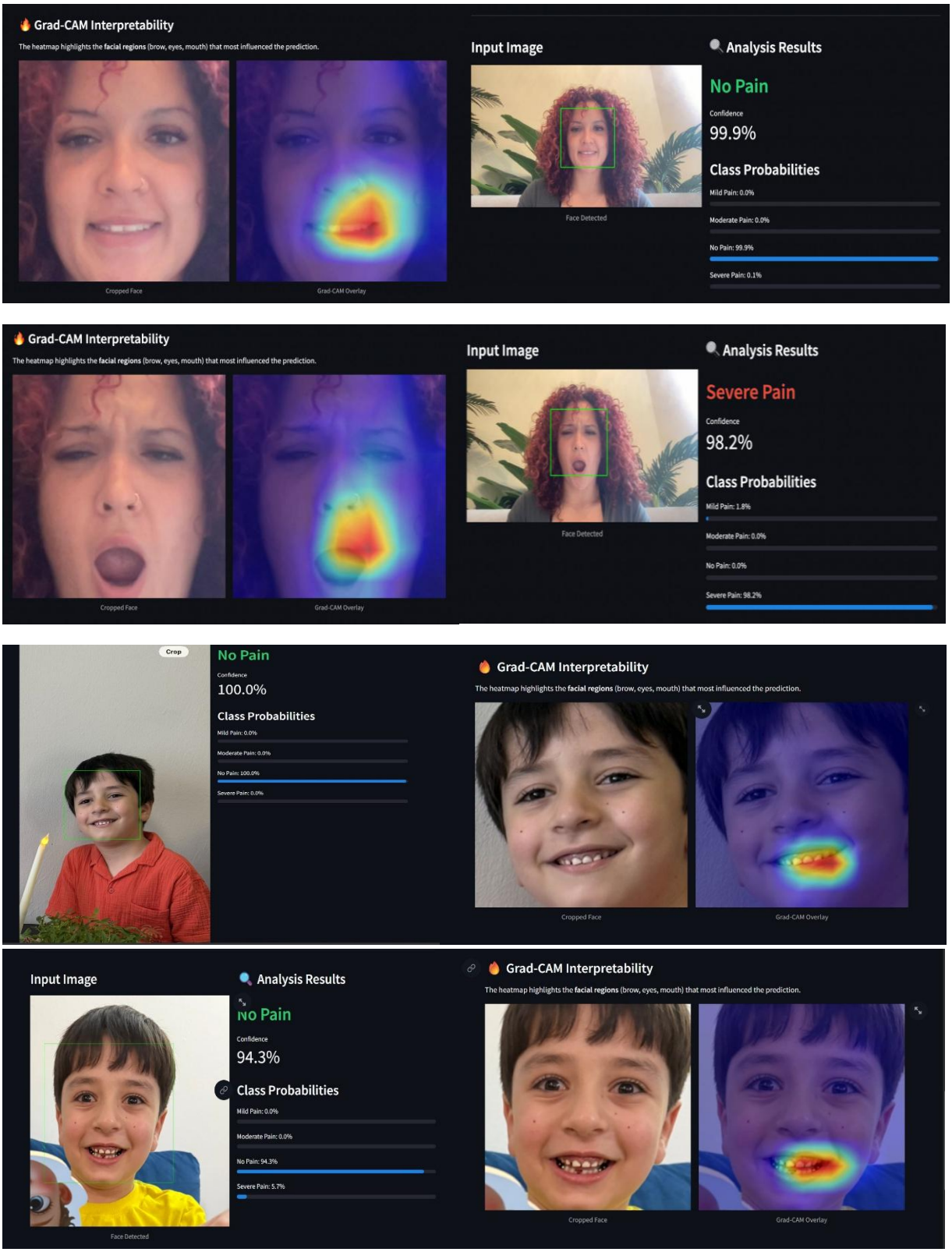


Figure 11: Example predictions and Grad-CAM visualizations from ResNet50 model

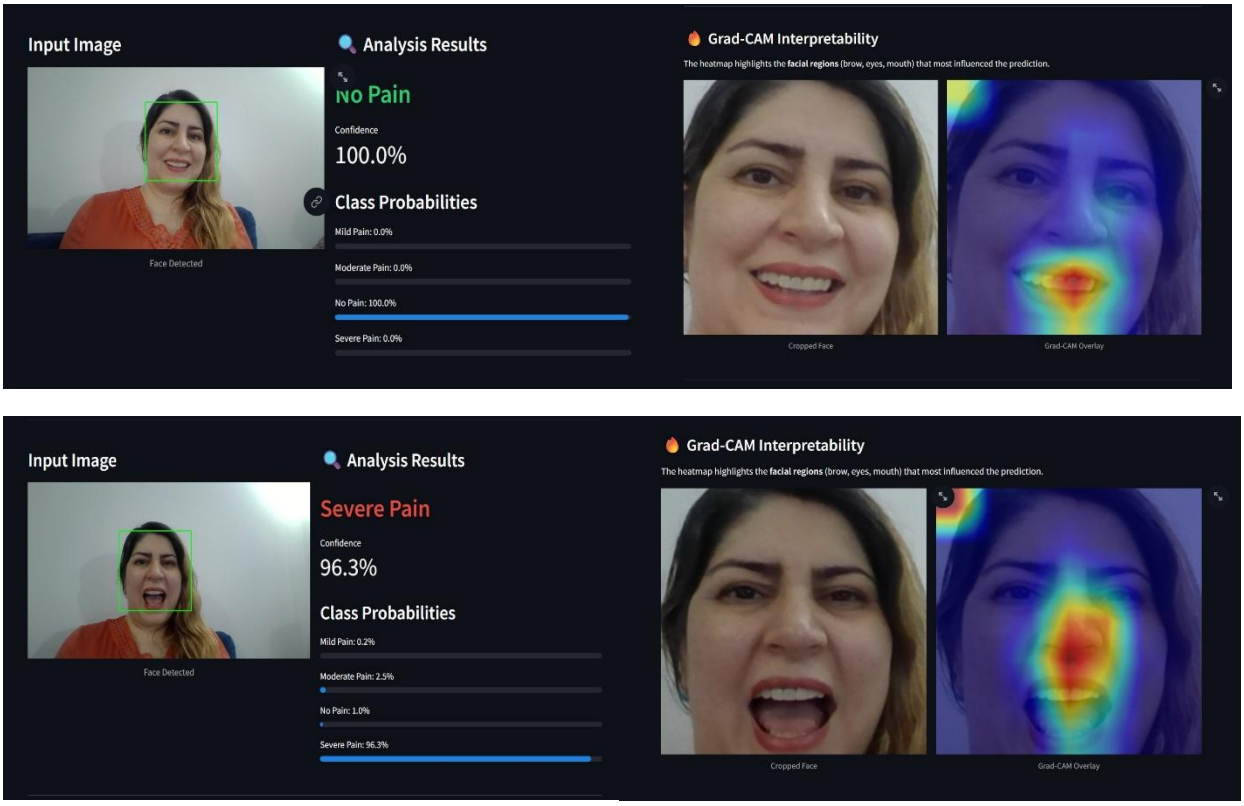
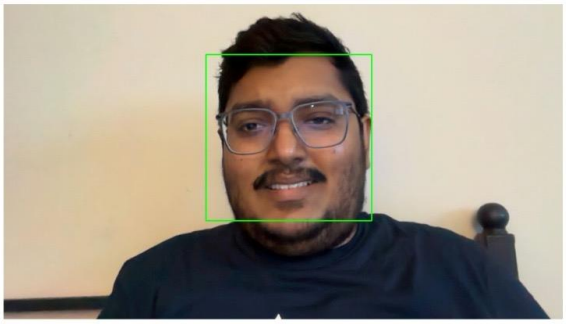


Figure 12: Example predictions and Grad-CAM visualizations from EfficientNet model

Input Image



Face Detected

🔍 Analysis Results

No Pain

Confidence

99.0%

Class Probabilities

Mild Pain: 0.2%

Moderate Pain: 0.1%

No Pain: 99.0%

Severe Pain: 0.8%

🔥 Grad-CAM Interpretability

Dual-branch Grad-CAM: left column = full-face encoder (ResNet-50), right column = mouth-crop encoder (ResNet-18). Red = high activation → what each branch focuses on.

Full Face

Grad-CAM: Full-Face Branch

Mouth Crop

Grad-CAM: Mouth Branch

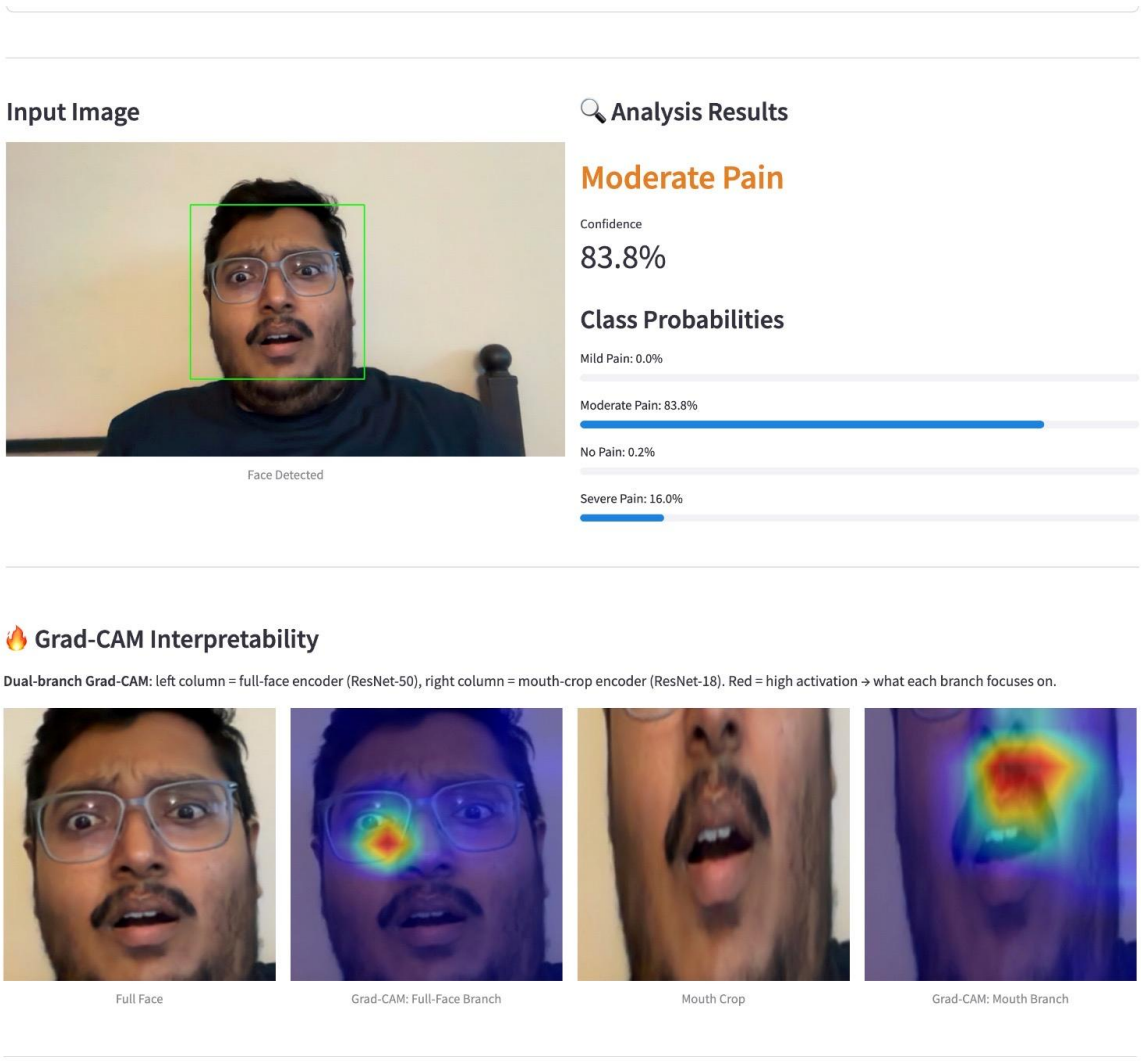


Figure 13: Example predictions and Grad-CAM visualizations from ResNet50 model

10.5 Key Observations

From the above results, several important observations can be made:

- Transfer learning models significantly outperform the baseline CNN
- VGG16 achieves the best overall performance
- Mild and moderate classes are the most difficult to distinguish
- The dual-input model improves stability and interpretability
- Grad-CAM confirms that models focus on meaningful facial regions

11. Discussion

The experimental results demonstrate that model architecture plays a critical role in the performance of facial pain classification. The baseline custom CNN achieved relatively low performance, with an F1-score of approximately 26%, indicating that training a model from scratch is insufficient for capturing the complex and subtle facial patterns associated with pain intensity. This is further supported by the confusion matrix, which shows widespread misclassification across all classes.

In contrast, transfer learning models such as VGG16, ResNet50, and EfficientNet-B3 significantly improved performance, achieving F1-scores above 77%. These models benefit from pretrained feature representations learned from large-scale image datasets, enabling them to extract more meaningful and discriminative features. Among all models, VGG16 achieved the best overall performance, suggesting that its architecture is particularly effective for capturing hierarchical facial features relevant to pain expression.

Despite these improvements, the confusion matrices reveal that classification errors are not uniformly distributed across classes. The models consistently perform well on the *severe* and *no pain* categories, where facial expressions are more distinct. However, there is noticeable confusion between the *mild* and *moderate* classes. This reflects the inherent difficulty of the task, as these categories involve subtle and overlapping facial features that are challenging to distinguish, even for human observers.

The analysis of training curves provides further insight into model behavior. While all transfer learning models show strong convergence, there is a consistent gap between training and validation performance, particularly in EfficientNet-B3 and ResNet50. This suggests the presence of mild overfitting, where the model learns training data patterns more effectively than generalizable features. Techniques such as stronger regularization, data augmentation, or early stopping could help mitigate this issue.

The dual-input model represents an important extension of the standard architecture by incorporating both full-face and mouth-region inputs. The training curves indicate more stable validation performance compared to single-input models, suggesting improved generalization. Additionally, Grad-CAM visualizations confirm that the model focuses more strongly on the mouth region, which is a key area for pain expression.

This suggests that incorporating localized facial information provides additional discriminative power, particularly for distinguishing between subtle pain levels such as mild and moderate.

Grad-CAM analysis across all models further supports the validity of the learned features. The visualizations consistently highlight regions around the eyes and mouth, which are known to be critical for expressing pain. This alignment between model attention and human understanding increases confidence in the model's predictions.

Overall, the results highlight the effectiveness of transfer learning for this task and demonstrate that incorporating localized facial information can further improve model performance. However, the reliance on relabeled emotion data rather than clinically annotated pain data remains a key limitation. This may restrict the model's ability to fully capture true pain-specific features and affects its applicability in real-world clinical settings.

12. Model Interpretability

To improve transparency, Grad-CAM can be used to visualize which regions of the face contribute most to the model's predictions. These visualizations often highlight areas such as the eyes and mouth, confirming that the model focuses on meaningful features.

13. Deployment

The trained model is designed for deployment using a web-based interface. A Streamlit application serves as the frontend, while model inference is handled either directly or through a Flask backend. The system can be hosted on Hugging Face Spaces, allowing users to upload images and receive predictions in real time.

14. Limitations

Despite promising results, several limitations remain. The dataset is not clinically labeled for pain, which limits the real-world applicability of the model. Additionally, the classification task is inherently difficult due to subtle differences between classes.

Variations in lighting, pose, and image quality may also affect performance. Furthermore, pain is a complex phenomenon that may not be fully captured by facial expressions alone.

15. Conclusion

This project demonstrates that deep learning models can effectively classify facial pain intensity. While the baseline CNN provides limited performance, transfer learning models significantly improve results. The best-performing model, VGG16, achieves an F1-score of 78.80%.

The dual-input architecture further enhances the system by combining global and local features, highlighting the importance of domain-specific design. Overall, this work provides a strong and scalable foundation for future research in automated pain assessment systems.

16. Future Work

Future work can focus on improving both the dataset and model architecture to enhance performance and real-world applicability. One important direction is the use of clinically labeled pain datasets, which would allow the model to learn true pain-specific features rather than approximations based on emotional expressions.

Additionally, incorporating temporal information from video data could help capture dynamic facial changes over time, which are important for accurate pain assessment. Future models could also explore attention mechanisms or transformer-based architectures to better focus on critical facial regions.

Another promising direction is the integration of multimodal data, such as voice, physiological signals, or behavioral cues, to provide a more comprehensive understanding of patient pain. Finally, further work can focus on improving model robustness and fairness to ensure reliable performance across diverse populations and real-world conditions.

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